Intranet site by just typing in the address of our VM. On the other hand, traffic to the public part of our Web server will be handled by Pound, a reverse proxy, which will filter our inbound traffic and automatically forward it to port 8888.

Our problem is this: We just cannot let our Intranet server be forwarded by our reverse proxy. If we do that, one fine day Google will decide to index our website, and it'll come up with all sorts of nasty trade secrets. *But our Web mail access was supposed to be via our Intranet site.* No, no! We cannot do that. What we can do, however, is to leave our Intranet site as it is and create a separate (a third) socket bind for our Mail server. If you want to know the sort of problems Google causes, go to the following site, which I found by searching for it in Google: *www.partners.org/email*

Anyway, create this bind:

```
$SERVER["socket"] == ":9999"{
    server.document-root        = "/Web/Webmail"
}
```

Of course, create the directories and test the stuff with our PHPINFO tactic.

That finishes the Web server part. Now for our reverse proxy.

Part 2: Pound for Pound

We need to set up a reverse proxy (RP) to forward all HTTP requests from our server to our Web VM. We use a RP called Pound to do it.

Why do we need a RP? Putting all our VMs directly on the Internet would mean that everyone can access our Web, database and mail servers. That would be suicide. C'mon, once someone guesses the root password for our DB server, we're finished. We are not going to put anything directly on the Internet. What we'll do instead is create reverse proxies to filter wanted traffic, and also abstractise all our VMs and show ONE computer to the whole world. Do you really think *www.google.com* is just one server?

But before we set up our reverse proxy, we need to alter our networking set-up.

Part 2½: Virtual distributed Ethernet

Now push your CentOS VM into the background and come to your Ubuntu. Launch a terminal and execute the following:

```
# apt-get install vde2 bridge-utils
#  echo "tun" >> /etc/modules
# modprobe tun
# tunctl -t tap0
# vde_switch -tap tap0 -daemon
# chmod -R a+rwx /var/run/vde.ctl
# brctl addbr br0
# ifconfig tap0 0.0.0.0 promisc
# brctl addif br0 tap0
# brctl addif br0 eth1
# ifconfig br0 10.111.111.254 netmask 255.255.255.0 up
```

We're done...almost! All we need to do now is tweak DNSMASQ a bit. Open up the DNSMASQ config file, and make a few changes:

1. Add one *except* line:
   ```
   except-interface eth1
   ```
2. Add one *interface* line. This should not be required, but it's better to be safe:
   ```
   interface=br0
   ```
3. And a CNAME record (You'll realise later why this is required)
   ```
   cname=webmail.officenetwork.local,web.officenetwork.local
   ```

Re-start DNSMASQ. It'll now run exactly as we want it to.

Shut down the CentOS VM, and then in the VBox GUI, open Networking Settings. You had attached your machine to eth1; now separate it from eth1 and attach it to tap0. Save your settings, and re-launch the VM. Pretty soon, you'll see it boot up and get all its networking information from our server over the Virtual Network.

Back to Part 2: The Pound reverse proxy

Anyway, we need to get pounding unless we want to keep our website one great secret. To start with, query the eth0 IP address:

```
# ifconfig eth0
```

Now get running. Install Pound:

```
# apt-get install pound
```

That should do the trick. Pound is now installed. All we need to do is configure it. *Why is configuring something always a pain in the posterior?*

Anyway, open up the Pound configuration file (*/etc/pound/pound.cfg*). To start with, edit what is re-usable:

1. Change the user and group. For this series, I'm doing everything as the root. You can select anything as per your own security model.
2. I personally prefer Apache type logs. To get them, change LogLevel to 3.

Now we need to set up the actual reverse proxy part. Delete the ListenHTTP segment at the end of the file, and start from scratch. I'll give you the entire configuration in the form of a snippet as below, but you need to change the domain names as per your own domain registrations. Specifically, I'm using *sample.co.cc*, and *officenetwork.local* but you need to change the sample to your own registered domain.

Also, you got the eth0 address earlier using the ifconfig command. It should have been 192.168.1.2, but it could be anything. Just take note of that too. And make sure that the address is different for eth0 and eth1.

Use the following snippet: